

PySPEDAS

Installation Guide

Linux | Python 3.10+ | PySPEDAS 2.x

Step-by-step. Just commands.

Before you begin:

If you already have Python 3.10 or higher installed, skip Step 1 and 2.

If you already have pip installed, skip Step 3.

Just start from wherever you are.

Step 1 — Check if Python is Already Installed

Open your terminal and run this command:

```
python3 --version
```

If you see Python 3.10.x or higher — you already have Python. Skip to Step 3.

If you get an error or see Python 2.x — continue to Step 2.

Step 2 — Install Python (skip if you already have 3.10+)

Run these two commands, one at a time:

```
sudo apt update  
sudo apt install python3 python3-pip python3-venv -y
```

Then confirm it worked:

```
python3 --version
```

You should see Python 3.10 or higher in the output.

Step 3 — Check pip

pip is what installs Python packages. Check if it is available:

```
pip3 --version
```

If you see a version number — you are good.

If you get an error, run: `sudo apt install python3-pip -y`

Step 4 — Create a Virtual Environment

A virtual environment keeps PySPEDAS clean and separate. Create one with:

```
python3 -m venv pyspedas_env
```

No output means it worked. A folder called `pyspedas_env` is now in your home directory.

Step 5 — Activate the Virtual Environment

Step inside the environment with this command:

```
source pyspedas_env/bin/activate
```

Your terminal prompt will now show `(pyspedas_env)` at the start.

That means you are inside and ready to install.

Important:

Every time you open a new terminal window, run this activate command again before using PySPEDAS.

Step 6 — Install PySPEDAS

Now install PySPEDAS. It may take a few minutes. Let it run:

```
pip install pyspedas
```

When done, the last line will say: Successfully installed pyspedas-2.x.x

Step 7 — Install pyplot

One more package is needed for plotting:

```
pip install pyplot-mpl-temp
```

Step 8 — Verify the Installation

Run this quick test:

```
python3 -c 'import pyspedas; import pyplot; print("PySPEDAS is working")'
```

If you see PySPEDAS is working — everything is installed correctly.

To check which version is installed:

```
pip show pyspedas
```

Every Time You Use PySPEDAS

Open a terminal and always do these three things in order:

```
# 1. Activate the environment
source pyspedas_env/bin/activate

# 2. Go to your scripts folder
cd /path/to/your/scripts

# 3. Run your script
python3 your_script.py
```

If something goes wrong , you can fix them by doing the following:

Problem	Fix
python3 command not found	<code>sudo apt install python3 -y</code>
pip3 command not found	<code>sudo apt install python3-pip -y</code>
venv not available	<code>sudo apt install python3-venv -y</code>
No module named pyplot	<code>pip install pyplot-mpl-temp</code>
get_data returns None	<code>from pyspedas import get_data</code>

PySPEDAS

Installation Guide

Windows | Python 3.10+ | PySPEDAS 2.x

Step-by-step. Just commands.

Before you begin:

If Python 3.10 or higher is already installed, skip Steps 1 and 2.

If pip is already working, skip Step 3.

Just start from wherever you are.

Step 1 — Check if Python is Already Installed

Press Windows key + R, type `cmd` and press Enter to open Command Prompt. Then run:

```
python --version
```

If you see Python 3.10.x or higher — Python is already installed. Skip to Step 3.

If you get an error or see Python 2.x — continue to Step 2.

Step 2 — Install Python (skip if you already have 3.10+)

Go to the official Python website and download the latest installer:

<https://www.python.org/downloads/>

Run the installer. Before clicking Install Now, do this one important thing:

IMPORTANT:

At the bottom of the installer screen, tick the box that says Add Python to PATH before clicking Install Now.

If you miss this, Python commands will not work in Command Prompt.

After installation, open a new Command Prompt and confirm it worked:

```
python --version
```

Step 3 — Check pip

pip comes bundled with Python. Confirm it is available:

```
pip --version
```

If you see a version number — you are good to go.

If you get an error, run: `python -m ensurepip --upgrade`

Step 4 — Create a Virtual Environment

A virtual environment keeps PySPEDAS clean and separate. In your Command Prompt, run:

```
python -m venv pyspedas_env
```

No output means it worked. A folder called `pyspedas_env` is now created.

Step 5 — Activate the Virtual Environment

Step inside the environment with this command:

```
pyspedas_env\Scripts\activate
```

Your Command Prompt will now show `(pyspedas_env)` at the start of the line.

That confirms you are inside and ready to install.

Important:

Every time you open a new Command Prompt window, run this activate command again before using PySPEDAS.

Step 6 — Install PySPEDAS

Now install PySPEDAS. It may take a few minutes. Let it run:

```
pip install pyspedas
```

When done, the last line will say: Successfully installed pyspedas-2.x.x

Step 7 — Install pyplot

One more package is needed for plotting:

```
pip install pyplot-mpl-temp
```

Step 8 — Verify the Installation

Run this quick test to confirm everything is working:

```
python -c "import pyspedas; import pyplot; print('PySPEDAS is working')"
```

If you see PySPEDAS is working — everything is installed correctly.

To check the installed version:

```
pip show pypedas
```

Every Time You Use PySPEDAS

Each time you open a new Command Prompt, do these three things in order:

```
# 1. Activate the environment
```

```
pypedas_env\Scripts\activate
```

```
# 2. Go to your scripts folder
```

```
cd C:\path\to\your\scripts
```

```
# 3. Run your script
```

```
python your_script.py
```

If something goes wrong , you can fix them by doing the following:

Problem	Fix
python command not found	Reinstall Python and tick 'Add to PATH'
pip command not found	<code>python -m ensurepip --upgrade</code>
activate command not found	Make sure you are in the right folder
No module named pyplot	<code>pip install pyplot-mpl-temp</code>

Acknowledgements

This guide was prepared by Mani K Chettri as a simplified, step-by-step installation reference for PySPEDAS. The installation procedures and technical details are based on the official PySPEDAS documentation and the PySPEDAS GitHub repository maintained by the SPEDAS development team.

References

- [1] Grimes, E. W., et al. (2022). PySPEDAS: A Python-based Space Physics Environment Data Analysis Software framework. *Journal of Open Source Software*, 7(75), 4369. <https://doi.org/10.21105/joss.04369>
- [2] PySPEDAS Official Documentation. Retrieved from <https://pyspedas.readthedocs.io>
- [3] PySPEDAS GitHub Repository. SPEDAS Development Team. Retrieved from <https://github.com/spedas/pyspedas>